

ShopTrail

Mobile Shopping App — Analytics Dataset Documentation

ShopTrail is a fictional mobile-first e-commerce application selling everyday lifestyle products across Electronics, Fashion, Home, Beauty, Sports, and Books. This dataset simulates the app's user base, in-app behavior, purchase transactions, product catalog, marketing campaign performance, and supporting reference data, structured as a complete star schema for a Power BI analytics project covering user engagement, revenue, and marketing ROI.

Table	Rows	Cols	Type	Purpose
Users	10,000	21	Dimension	User profiles & demographics
Countries	36	7	Dimension	Geo lookup for map visuals
Products	500	11	Dimension	Product catalog
TrafficSources	5	7	Dimension	Acquisition channel details
SubscriptionPlans	4	9	Dimension	Plan pricing & features
PaymentMethods	7	5	Dimension	Payment type details
Screens	8	5	Dimension	App screen metadata
Sessions	100,000	16	Fact	App login sessions
AppEvents	500,000	14	Fact	In-app interaction events
Purchases	20,000	18	Fact	Order transactions
Campaigns	120	15	Fact	Marketing campaign performance

11 tables (7 dimension + 4 fact) | ~630,000 total records | Format: .xlsx / .csv | Built for: Power BI

Data Model Overview

The dataset follows a star-schema design. **Users, Countries, Products, TrafficSources, SubscriptionPlans, PaymentMethods,** and **Screens** are dimension tables. **Sessions, AppEvents, Purchases,** and **Campaigns** are fact tables. Every table connects into a single model — there are no orphan tables.

Relationships & Cardinality

From (many)	Key	To (one)	Cardinality
Users	CountryID	Countries	Many-to-One
Users	SubscriptionPlan	SubscriptionPlans	Many-to-One
Users	SignupChannel	TrafficSources	Many-to-One
Users	AcquisitionCampaignID	Campaigns	Many-to-One
Sessions	UserID	Users	Many-to-One
AppEvents	SessionID	Sessions	Many-to-One
AppEvents	UserID	Users	Many-to-One
AppEvents	ScreenName	Screens	Many-to-One
Purchases	UserID	Users	Many-to-One
Purchases	ProductID	Products	Many-to-One
Purchases	PaymentMethod	PaymentMethods	Many-to-One

All relationships are one-to-many, flowing from a dimension table (the “one” side) to a fact table (the “many” side) — the standard, correct pattern for a star schema. In Power BI, set filter direction to **single, dimension** → **fact**. Note that **Users.AcquisitionCampaignID** is blank for Organic and Referral signups, since those users were not driven by a paid campaign — this is intentional, not missing data. No many-to-many relationships exist in this model.

Data Dictionary

Dimension Tables

1. Users (10,000 rows)

Registered ShopTrail users with demographic, location, and account details.

Column	Type	Description
UserID	Integer	Unique identifier for each user (primary key)
UserName / Email / Phone	Text	User contact and identity details
Gender / Age / AgeGroup	Mixed	Demographic attributes
CountryID	Integer	Foreign key to Countries table
Country / State / City	Text	User's location
Latitude / Longitude	Decimal	Coordinates for map visuals
SignupDate	Date	Date the user registered
SignupMethod	Text	Email / Google / Facebook / Apple / Phone
SignupChannel	Text	Foreign key to TrafficSources (by name)
PreferredLanguage	Text	User's app language preference
SubscriptionPlan	Text	Foreign key to SubscriptionPlans (by name)
UserStatus	Text	Active / Inactive / Suspended / Deleted
LastLogin	Date	Most recent login date
AcquisitionCampaignID	Integer	Foreign key to Campaigns; blank for Organic/Referral users

2. Countries (36 rows)

Geographic lookup table supporting Power BI map visuals.

Column	Type	Description
CountryID	Integer	Unique country identifier (primary key)
Country / State / City	Text	Location hierarchy
Latitude / Longitude	Decimal	Geographic coordinates
TimeZone	Text	Local time zone (GMT offset)

3. Products (500 rows)

ShopTrail's product catalog across six categories.

Column	Type	Description
ProductID	Integer	Unique product identifier (primary key)
ProductName / Category / SKU	Text	Product identity and classification
Supplier	Text	Vendor supplying the product
CostPrice / SellingPrice	Decimal	Cost and retail pricing
Discount	Integer	Standard discount percentage
StockQuantity	Integer	Units currently in stock
Warehouse	Text	Fulfillment warehouse location
LaunchDate	Date	Date product was added to the catalog

4. TrafficSources (5 rows)

Acquisition channel reference table describing how users found ShopTrail.

Column	Type	Description
TrafficSourceID	Integer	Unique identifier (primary key)
SourceName	Text	Organic / Paid Ads / Referral / Social Media / Influencer
SourceType	Text	Owned / Paid / Earned
Channel	Text	Specific channel detail (e.g., Google Ads)
Description	Text	Explanation of the source
CostModel	Text	Free / CPC / CPM / Flat Fee / Flat Reward
AvgCostPerAcquisition	Decimal	Average cost to acquire one user via this source

5. SubscriptionPlans (4 rows)

Pricing and feature details for each ShopTrail subscription tier.

Column	Type	Description
PlanID	Integer	Unique identifier (primary key)
PlanName	Text	Free / Basic / Premium / Premium+
MonthlyPrice / AnnualPrice	Decimal	Subscription cost
TrialDays	Integer	Free trial length
AdFree	Boolean	Whether ads are removed on this plan
MaxDevices	Integer	Devices allowed per account
FreeShipping	Boolean	Whether shipping is free on this plan

Column	Type	Description
Features	Text	Summary of plan benefits

6. PaymentMethods (7 rows)

Reference table describing each payment option available at checkout.

Column	Type	Description
PaymentMethodID	Integer	Unique identifier (primary key)
MethodName	Text	Credit Card / Debit Card / UPI / PayPal / Net Banking / Mobile Wallet / COD
MethodType	Text	Card / Digital Wallet / Bank Transfer / Cash
IsOnline	Boolean	Whether the method is processed online
ProcessingFeePct	Decimal	Processing fee as a percentage

7. Screens (8 rows)

Reference table describing each screen tracked in AppEvents.

Column	Type	Description
ScreenID	Integer	Unique identifier (primary key)
ScreenName	Text	Home / ProductDetail / Cart / Checkout / Profile / Search / Settings / Wishlist
ScreenCategory	Text	Core / Commerce / Account / Discovery
IsCoreScreen	Boolean	Whether this is a primary navigation screen
Description	Text	Purpose of the screen

Fact Tables

8. Sessions (100,000 rows)

Individual app login sessions capturing device and connectivity context.

Column	Type	Description
SessionID	Integer	Unique session identifier (primary key)
UserID	Integer	Foreign key to Users table
SessionDate / LoginTime / LogoutTime	Datetime	When the session occurred
DeviceType / DeviceBrand	Text	Device used for the session
Browser / OS / AppVersion	Text	Technical environment
NetworkType	Text	WiFi / 4G / 5G / 3G
SessionStatus	Text	Completed / Dropped / Timeout
IsFirstSession	Boolean	Whether this was the user's first-ever session
Country / City	Text	Session location
Duration	Decimal	Session length in minutes

9. AppEvents (500,000 rows)

Granular in-app interaction events for behavioral and funnel analysis.

Column	Type	Description
EventID	Integer	Unique event identifier (primary key)
SessionID / UserID	Integer	Foreign keys to Sessions and Users
EventCategory / EventLabel	Text	Type and nature of the event
ScreenName	Text	Foreign key to Screens table
ScreenCategory	Text	Core / Commerce / Account / Discovery
EventValue / TimeSpent	Decimal	Numeric value and time spent
SearchKeyword / ButtonClicked	Text	Contextual interaction detail
ScrollDepth	Integer	Percentage of screen scrolled (0-100)
PurchaseFlag	Integer	1 if event led to a purchase, else 0
EventTimestamp	Datetime	Exact date and time of the event

10. Purchases (20,000 rows)

Completed and in-progress order transactions on ShopTrail.

Column	Type	Description
PurchaseID	Integer	Unique order identifier (primary key)
UserID / ProductID	Integer	Foreign keys to Users and Products
Quantity / PurchaseDate	Mixed	Order size and date
UnitPrice / Subtotal	Decimal	Pricing at time of order
CouponCode	Text	Discount code applied, if any
OrderStatus	Text	Delivered / Shipped / Processing / Cancelled / Returned
ShippingCost / Tax	Decimal	Additional order charges
DeliveryDays / DeliveryPartner	Mixed	Fulfillment details
PaymentMethod	Text	Foreign key to PaymentMethods table
PaymentStatus	Text	Paid / Pending / Failed / Refunded
ReturnStatus	Text	Refunded / Exchange / In Transit / Not Applicable
CostPrice / Profit	Decimal	Order-level cost and profit

11. Campaigns (120 rows)

Marketing campaign performance across platforms and channels.

Column	Type	Description
CampaignID	Integer	Unique campaign identifier (primary key); referenced by Users.AcquisitionCampaignID
CampaignName / CampaignType	Text	Campaign identity and category
Platform	Text	Facebook / Instagram / Google Ads / YouTube / TikTok / Email
StartDate / EndDate	Date	Campaign flight dates
Budget / Spend	Decimal	Planned vs actual spend
Impressions / Clicks / CTR	Mixed	Reach and engagement metrics
Installs / Conversions	Integer	Attributed installs and conversions
Revenue / ROI	Decimal	Attributed revenue and return on investment

Suggested Power BI Analysis

- User Engagement: sessions per user, average session duration, device/OS breakdown, first-time vs returning users
- Behavioral Funnel: screen-to-screen navigation flow using Screens, drop-off points, scroll depth vs conversion
- Revenue & Profitability: revenue and profit by category, top-selling products, discount impact on margin
- Geographic Insights: user distribution and revenue by country/state/city using the Countries map table
- Marketing ROI & Attribution: campaign spend vs revenue, CTR/conversion by platform, and full-funnel view from TrafficSources and AcquisitionCampaignID through to Purchases
- Subscription Analysis: plan distribution, revenue potential by plan using SubscriptionPlans pricing
- Payment Behavior: preferred payment methods by region or plan, processing fee impact using PaymentMethods
- Retention & Lifecycle: signup channel effectiveness, subscription plan distribution, churn signals via UserStatus

Note: ShopTrail is a fictional app created for portfolio purposes. All data in this dataset is synthetically generated and does not represent real users, transactions, or companies.